75 Allogeneic Hematopoietic Cell Transplantation in Pediatric MDS Including Refractory Cytopenia of Childhood and in Juvenile Myelomonocytic Leukemia

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75.1 Introduction

Most pediatric patients with MDS or juvenile myelomonocytic leukemia (JMML) can be cured by allogeneic HCT. Since MDS and JMML are associated with heterogeneous germline genetic conditions and/or somatic oncogenic mutations, specific attention to their subtypes is crucial.

75.2 Pediatric MDS Including Refractory Cytopenia

75.2.1 Classification

Pediatric MDS can be associated with germline predisposition, related to cytotoxic or immunomodulatory therapy, or occurs as de novo disease. The term primary MDS summarizes de novo MDS and MDS in germline predisposition other than the classical inherited BMF disorders. MDS can present as MDS-EB with 2–19% blasts in PB and/or 5-19% blasts in BM. Occasionally, disease with 20–30% blasts lacks clinical features of acute leukemia and behaves more like MDS than AML. In approximately 80% of MDS cases, the BM is hypocellular. Patients with a hypocellular BM and normal blast count may present with a pattern of refractory cytopenia of childhood (RCC), a well-recognized type of BM failure in children with persistent cytopenia and evidence of dysplasia (Arber et al. 2022).

75.2.2 Germline Predisposition and Monosomy 7

GATA2 deficiency and SAMD9/SAMD9L syndrome emerged as the most frequent hereditary cause of primary pediatric MDS with a prevalence of 8% and 7%, respectively (Sahoo et al. 2021). Half of all cases of primary MDS with monosomy 7 arise from germline *SAMD9/9 L* or *GATA2* mutations. In SAMD9/SAMD9L syndrome, somatic genetic rescue events are frequent, and spontaneous loss of monosomy 7 with hematologic recovery can occur in children <5 years of age. Germline mutations in *RUNX1*, *ETV6*, *ANKRD26* and *ERCC6L2* may also give rise to MDS in young individuals.

Considerations for HCT in predisposition syndromes listed above are generally no different from those for HCT in wildtype conditions. Retrospective analyses of EWOG-MDS indicated similar outcome of SAMD9/9 L syndrome, GATA2 deficiency syndrome and wildtype when stratified according to blast count, karyotype and BM cellularity (Sahoo et al. 2021; Bortnick et al. 2021). The acute post-transplant course may,

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however, be complicated by syndrome-related comorbidities (Ahmed et al. 2019). In addition, potential family donors need to be evaluated for presence of the respective underlying predisposition.

75.2.3 MDS-EB with *UBTF-TD*, Role of Cytoreductive Therapy Prior to HCT

Tandem duplication in upstream binding transcription factor (UBTF) are noted in close to a third of patient with "de novo" MDS-EB (Erlacher et al. 2022). UBTF-TD is associated with normal karyotype or trisomy 8, poor response to AML-type chemotherapy, and inferior outcome after HCT compared to wildtype.

While the role of intensive chemotherapy prior to HCT remains unknown, regimens like venetoclax and hypomethylating agents may bridge patients with increasing blast count to HCT (Masetti et al. 2023). In germline conditions with immunodeficiency (like GATA2 and SAMD9/SAMD9L syndrome), intensive chemotherapy prior to HCT is to be avoided whenever possible.

75.2.4 HCT in Primary MDS with Normal Blast Count

MDS with monosomy 7 is at high risk of progression, and patients should be transplanted as soon as possible. For RCC with monosomy 7, del(7q) or ≥2 aberrations, MAC is recommended. EWOG-MDS currently advocates a TREO-based regimen, which results in prompt initial engraftment with a low incidence of secondary graft failure and an OS of approx. 90% (see https://ewog-mds.org). Historical data with BU/CY demonstrate an OS of 75% with NRM being the major cause of treatment failure.

In the absence of monosomy 7, RCC patients with mild cytopenia (not transfusion dependent for RBC and/or platelets, absolute neutrophil count > 1×10^9/L) may have a stable course of disease and therefore qualify for a watch-and-observe strategy. For patients with more pronounced cytopenia treatment is stratified according to cellularity. In normo- or hypercellular BM a MAC regimen like that described for monosomy 7 can be utilized irrespective of karyotype. In hypocellular BM and normal karyotype, HCT with RIC is the treatment of choice. HCT with a preparative regimen of TT/FLU resulted in an OS of 94% and EFS of 88% (Strahm et al. 2017). However, approx. 10% of patients experience primary and secondary graft failure requiring a stem cell boost and/or second HCT. Thus, EWOG-MDS currently recommends a preparative regimen of TREO/FLU resulting in an improved rate of engraftment (see https://ewog-mds.org). With a very low risk of disease recurrence, GVHD should be avoided, and BM is the preferred stem cell source combined with an effective GVHD prophylaxis.

75.2.5 HCT in Primary MDS-EB

In a large cohort, allo-HCT with full MAC consisting of the combination of BU/CY/MEL resulted in a probability of OS at 5 years of 63%, with TRM and relapse contributing equally to treatment failure (Strahm et al. 2011). Outcome for patients, who received a graft from an MSD or a UD matched for 9/10 or 10/10 HLA-loci is superimposable. Because patients ≥ 12 years of age had a high risk of NRM, EWOG-MDS recommends an intensified GVHD prophylaxis (CSA/MTX) for older patients transplanted from an MSD (see https://ewog-mds.org). Presence of a structurally complex karyotype is strongly associated with poor prognosis.

75.2.6 HCT in Therapy-Related MDS

In this heterogeneous patient population, OS in the presence of a structural complex karyotype and/or *TP53* mutation is ≤20%, for all other karyotypes/molecular subgroups approx. 50% (Kornemann et al. 2023).

75.3 Juvenile Myelomonocytic Leukemia

75.3.1 Clinical Features and Genetic Subtypes

JMML is a unique clonal hematopoietic disorder of early childhood. Splenomegaly, leukocytosis, monocytosis, and myeloid and/or erythroid precursors on PB smear are noted in close to all cases (Niemeyer and Flotho 2019). The pathobiology is characterized by constitutive activation of the RAS signal transduction pathway. Canonical RAS pathway mutations in the *PTPN11*, *NRAS*, *KRAS*, *NF1, CBL* genes are present in leukemic cells of more than 95% of patients and define genetically and clinically distinct subtypes.

75.3.2 Risk Factors and Indication of HCT

Risk factors for poor outcome are age ≥ 2 years, hemoglobin F ≥ 15%, DNA hypermethylation (cross-continental molecular classifier in high, intermediate, low methylation) and presence of secondary mutations (e.g., *SETBP1*, *JAK3*, other *RAS* pathway genes). Risk factors are strongly correlated and allow risk allocation within genetic subtypes.

75.3.3 Therapy Prior to HCT

For approximately 80% of patients with JMML early alloHCT is the therapy of choice (Table 75.1). In Europe, standard therapy prior to

Table 75.1 Characteristics of genetic subtypes

Genetic subtype		
PTPN11 (41%)	Presentation	Non-syndromic
	Mutation	Heterozygous, somatic gain of function mutation, broad overlap with Noonan syndrome with transient myeloproliferative disease → analysis of non-hematopoietic cells (hair bulbs, buccal swab, fibroblasts) required
	Risk profile	Most often high
	HCT	Indicated
NF1 (12%)	Presentation	≥ 6 café au lait spots not always present; higher platelet count, higher blast % and more often >5 years of age compared to other subtypes
	Mutation	*NF1* germline mutation with biallelic inactivation in hematopoietic cells
	Risk profile	Often high
	HCT	Indicated
NRAS (16%)	Presentation	Non-syndromic
	Mutation	Heterozygous somatic *NRAS* mutations at codons 12, 13, rarely 61
	Risk profile	very heterogeneous; in some low-risk patients spontaneous clinical regression in the absence of therapy
	HCT	Indicated for high-risk patients
KRAS (14%)	Presentation	Non-syndromic, median age 0.9 years
	Molecular	heterozygous, somatic mutations involving codon 12, 13, rarely 61; monosomy 7 more frequent than in other subtypes
	Risk profile	Most often low or intermediate (by methylation class)
	HCT	Indicated for most patients; in some children sustained clinical and molecular CR following azacitidine therapy
CBL (12%)	Presentation	± Noonan-like phenotype, occasional patient with grossly enlarged spleen, risk of vasculopathy (moyamoya, opticus arteritis, and major arteries)
	Mutation	*CBL* germline mutation with biallelic inactivation; occasionally heterozygous mutation only
	Risk profile	Generally low
	HCT	Rarely required, may be indicated for patients with vasculopathy

(continued)

HCT is azacitidine (FDA approved) with best responses in low- and intermediate-risk patients (Niemeyer et al. 2021). 6-MP and low dose ara-C are effective as well. Patients with very aggressive disease (platelet transfusion dependent, pulmonary insufficiency) may benefit from high-dose Ara-C with FLU. MEK-inhibition (e.g., trametenib) alone may not result in significant responses (Stieglitz et al. 2021). Splenectomy prior to HCT improves neither engraftment nor outcome following HCT and is generally not recommended.

75.3.4 HCT and Outcome

MAC with BU/CY/MEL is standard conditioning for HCT in JMML from MSD and MUD (Locatelli 2005; Dvorak et al. 2018). Recommended stem cell source is BM. TRM varies among genetic subgroups (*PTPN11*-mut. 5%, *NF1*-mut. 27%). Relapse is the most important cause of failure. DFS and RI for patients (all genetic subtypes) with HbF ≤ 15% is 71% and 14%, respectively, and for HbF > 42% and 49%, respectively (EWOG-MDS unpublished). In high-risk disease, early tapering of immunosuppressive therapy in the absence of grade II-IV acute GVHD and steroid treatment is recommended (start at day +40, discontinuation between day +60 and +90). For patients with *PTPN11*-mutated JMML and a high risk of relapse, post engraftment therapy with azacitidine and DLI is currently being piloted. For children lacking a HLA-compatible relative, UBCT (Locatelli and Strahm 2018) or haploidentical HCT with T-cell depletion (i.e., TCRalpha/beta/CD19 or PTCY a potential option.

Key Points

- Outcome of HCT in MDS associated with GATA2 deficiency or SAMD9/SAMD9L syndrome is similar to that of wildtype when stratified according to blast count, karyotype, and BM cellularity.
- MDS-EB and MDS with monosomy 7 require an MAC, whereas hypocellular RCC without monosomy 7, del(7q) or ≥2 aberration can successfully be transplanted with a less intensive regimen.
- Tandem duplications in *UBTF* are noted in approximately one-third of pediatric patients with de novo MDS-EB. They are associated with poor response to intensive AML-type therapy and inferior outcome following MAC HCT.
- Molecular alterations in *PTPN11*, *NRAS*, *KRAS*, *NF1*, or *CBL* define clinically distinct JMML subtypes. Azacitidine is FDA approved for newly diagnosed JMML.
- Approx. 80% of children with JMML require HCT for cure. Standard preparative regimen is BU/CY/MEL.
- In JMML, risk factors for relapse following HCT (age > 2 years, HbF > 15%, presence of subclonal mutations, and high DNA methylation pattern) are closely related.

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